

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Solution Implementation Summary

The Emotion Detector platform was built entirely in Python using a modular, layered architecture. Below is a summary of how each major feature was implemented.

Module 1: Data Preprocessing (`src/preprocessing/`)

Dataset: dair-ai/emotion from HuggingFace Hub (16K training samples, 6 original classes remapped to 5)

Original Class	Mapped To
joy	Confident
sadness	Frustrated
anger	Frustrated
fear	Confused
love	Curious
surprise	Curious

Preprocessing Pipeline:

```
Raw Text  HTML Removal  URL/Mention Removal  Lowercase
Punctuation Removal  Stopword Removal (negation-preserving)
Lemmatization (WordNet)  Padded Integer Sequence
```

BERT Path: Minimal cleaning (preserve casing, punctuation context) BertTokenizer subword encoding

Module 2: BiLSTM Model (`src/models/bilstm/`)

```
# Model Architecture (Keras)
Input: (100,) # padded integer sequence
Embedding(10000, 128, input_length=100)
SpatialDropout1D(0.3)
Bidirectional(LSTM(128, dropout=0.3, recurrent_dropout=0.2, return_sequences=True))
Bidirectional(LSTM(64, dropout=0.3))
BatchNormalization()
Dense(64, activation='relu')
Dropout(0.4)
Dense(5, activation='softmax')
```

Training Config:

- Optimizer: Adam (lr=1e-3)
 - Loss: Categorical Cross-Entropy
 - Callbacks: EarlyStopping (patience=5), ReduceLROnPlateau, ModelCheckpoint
 - Class Weights: Computed from class distribution to handle imbalance
-

Module 3: BERT Model (`src/models/bert/`)

```
# Model Architecture (HuggingFace)
Base: bert-base-uncased (110M parameters)
BertModel (pretrained weights)
Dropout(0.3)
Linear(hidden_size=768, 5) # classification head
Softmax
```

Training Config:

- Trainer: HuggingFace `Trainer` API
 - Learning Rate: 2e-5 with linear warmup
 - Batch Size: 16 (train), 32 (eval)
 - Epochs: 5 with EarlyStopping (patience=2)
 - Mixed Precision: FP16 (GPU) / FP32 (CPU)
-

Module 4: Detection Engine (`src/detection/`)

```
# Weighted Ensemble
ensemble_probs = 0.40 * bilstm_probs + 0.60 * bert_probs

# Rule-Based Boost
if keywords_matched:
    ensemble_probs[matched_emotion] += 0.15
    ensemble_probs = ensemble_probs / ensemble_probs.sum() # renormalize

# Final Prediction
final_emotion = EMOTION_LABELS[np.argmax(ensemble_probs)]
final_confidence = np.max(ensemble_probs)

# Mixed Emotion Check
gap = ensemble_probs[0] - ensemble_probs[1] # top2 gap
is_mixed = gap < 0.15
```

Module 5: Gemini Guidance Generator (`src/gemini/`)

Prompt Template Structure:

```
You are an empathetic AI learning assistant.
A student who is feeling {EMOTION} (confidence: {CONFIDENCE}%) described:
"{STUDENT_TEXT}"

Provide a structured response with exactly these 5 sections:
1. Emotional Acknowledgement
2. Targeted Learning Resources
3. Recommended Study Techniques
4. Break and Recovery Suggestions
5. Words of Encouragement
```

Retry Logic: Exponential backoff (1s 2s 4s) on API rate limit errors

Module 6: Streamlit App (`app/`)

Page	Key Components
Home (Demo)	Text area, Detect button, Emotion card, Confidence bar chart, Guidance panel
Analytics	4 Plotly charts: Emotion distribution, Confidence trend, Session frequency, Mode
Model Comparison	BiLSTM vs BERT architecture cards, performance metrics

Page	Key Components
About	Project description, usage guide, tech stack

CSS Theme: Dark glassmorphism - #0F172A background, #6C63FF accent, Inter font, backdrop blur cards

Module 7: Analytics & Logging (`src/analytics/`)

```
# CSV Log Row
{
  "timestamp": "2026-07-19T10:30:00",
  "session_id": "sess_abc123",
  "final_emotion": "frustrated",
  "final_confidence": 0.87,
  "is_mixed": False,
  "models_agree": True,
  "rule_applied": True,
  "keywords_matched": {"frustrated": ["stuck", "can't"]},
  "guidance_excerpt": "I understand you're feeling frustrated..."
}
```

GitHub Repository

URL: https://github.com/Harsh-maker007/Smartbridge_project1

Stat	Value
Total Files	71+
Total Lines of Code	~5500
Test Coverage	25+ unit tests
Python Modules	20+
Commits	6+